

LeepTalent Level 3 End-point Assessment

World Happiness

Scenario Demonstration — Pack Introduction



Difficulty: 4 of 5 — Advanced

What this pack is

This is a practice Scenario Demonstration pack modelled on the City & Guilds 9665 Data Technician end-point assessment. It contains four parts: this introduction, the Scenario Demonstration brief, the data files, and a Learner Guide showing how the task can be approached and the results you should expect to reach.

What is being looked at

This scenario brings together three sources of different shapes and asks for correlation analysis across several variables. The areas examined are:

Area examined	Where it appears
Importing data from multiple formats (CSV, XLSX, document table)	Scenario 1, Task 1
Reshaping a transposed source (unpivoting columns to rows)	Scenario 1, Task 1 (population)
Validation: header consistency, duplicates, missing values	Scenario 1, Task 2
Appending two data sets and adding a Year column	Scenario 1, Task 3
Joining on a text key and reconciling name mismatches	Scenario 1, Task 4 (country names)
Year-over-year comparison with a calculated column	Scenario 2, Task 1
Aggregating and comparing against a benchmark / subgroup	Scenario 2, Tasks 2–3 (UK, G7)
Scatter plots and scale choice (log)	Scenario 2, Task 4
Correlation analysis across multiple variables	Scenario 2, Task 5

Why this is rated 4 stars

Two things lift this above the mid-tier. First, the population file is a World Bank export in a **transposed** layout — years down the rows, countries across the columns — which must be

unpivoted into a tidy format before it can be joined. Second, the join on country name carries a real **name-matching trap**: several countries are named differently in the two sources and will fail to match unless reconciled. It stops short of 5 stars because the analysis tasks, while broad, are clearly specified rather than fully open-ended.

Areas to watch

- The population file has **title rows** at the top and is **transposed** — unpivot it so each row is one country-year before joining.
- Country names differ across sources: **Swaziland** vs Eswatini, **Congo (Kinshasa)** vs Congo, Dem. Rep., **North Macedonia** vs Macedonia — a null population after the join flags a mismatch to fix.
- Add a **Year** column when appending 2018 and 2019, or the two years can't be told apart.
- Population sizes vary enormously — use a **log scale** on the scatter so smaller countries aren't squashed against the axis.
- Correlation shows association, not cause — report it carefully.

What's in the pack

- **Scenario Demonstration – World Happiness** (PDF) — the brief, in C&G format.
- **Happiness data 2018.csv** — 2018 happiness rankings.
- **Happiness data 2019.xlsx** — 2019 happiness rankings.
- **Population data.csv** — World Bank population data (transposed).
- **Annex A – Population Extract** — sample figures, embedded in the brief.
- **Learner Guide – World Happiness** (PDF) — method and expected results.